

Bioinformatics Tutorial

Evaluation Form — 150 points

EEB 603 Reproducible Science — Boise State University

Fall 2026

Student(s): _____

Chapter: _____

Date: _____

Evaluator: _____

1. Overview of the Assignment

This evaluation rubric is designed to assess student-developed bioinformatics tutorials created using **knitr/RMarkdown** in **RStudio**. Each tutorial should introduce key bioinformatics concepts, use real or simulated datasets, and demonstrate practical computational skills. Tutorials must be designed to fit within two laboratory sessions and meet the criteria specified in the assignment description.

2. Rubric

A. Introduction and Theoretical Background — 25 points

Criteria

- **Context & Relevance (10 pts):** Explains the scientific context and links to the scientific process and data life cycle.
- **Objectives (10 pts):** Clearly states tutorial aims and learning goals aligned with the chosen chapter.
- **Clarity & Cohesion (5 pts):** Introduction is well-structured, engaging, and sets up the tutorial logically.

Comments:

Score: _____ / 25

B. Learning Outcomes — 15 points

Criteria

- **Specificity & Actionability (10 pts):** Outcomes use measurable verbs (e.g., import, visualize, annotate, model).
- **Alignment (5 pts):** Outcomes align with activities, tools, and data used in the tutorial.

Comments:

Score: _____ / 15

C. R Package Requirements and Installation Instructions — 20 points

Criteria

- **Package Identification (10 pts):** Includes all necessary R packages and tools.
- **Installation & Dependencies (10 pts):** Provides installation commands, explanations, and troubleshooting notes.

Comments:

Score: _____ / 20

D. Dataset Description and Access — 15 points

Criteria

- **Dataset Explanation (10 pts):** Clearly introduces the dataset, origin, variables, and relevance to tutorial aims.
- **Access Instructions (5 pts):** Provides clear download instructions or embedded loading steps.

Comments:

Score: _____ / 15

E. Content Quality and Concept Explanations — 25 points

Criteria

- **Conceptual Accuracy (10 pts):** Concepts are explained correctly using appropriate detail.
- **Rationale & Stepwise Explanation (10 pts):** Clear reasoning, pseudocode, or workflow descriptions accompany each task.
- **Input/Output Documentation (5 pts):** Defines expected data types, formats, and outputs for each function or analysis step.

Comments:

Score: _____ / 25

F. R Code Quality and Annotation — 25 points

Criteria

- **Correctness of Code (10 pts):** Code runs without errors and produces expected outputs.
- **Commenting & Readability (10 pts):** Code is clearly annotated and easy to follow.
- **Reproducibility (5 pts):** Tutorial knits successfully; file paths and package calls are properly set.

Comments:

Score: _____ / 25

G. Structure, Organization, and Flow — 10 points

Criteria

- **Logical Flow (5 pts):** Tutorial moves naturally from theory → data → methods → exercises → interpretation.
- **User Experience (5 pts):** Sections are well-organized, readable, and accessible to peers.

Comments:

Score: _____ / 10

H. References and Attribution — 10 points

Criteria

- **Proper Citation (5 pts):** All sources (e.g., textbooks, manuals, online documentation) cited correctly.
- **Useful Resources (5 pts):** Provides links to package documentation, vignettes, or relevant literature.

Comments:

Score: _____ / 10

I. Professionalism and Submission Requirements — 5 points

Criteria

- **Submission Protocol (5 pts):**
 - Submitted 1 week before presentation
 - Uploaded to Google Drive
 - Email sent to class introducing the tutorial and resources

Comments:

Score: _____ / 5

3. Score Summary

Section	Points	Score
A. Introduction and Theoretical Background	25	
B. Learning Outcomes	15	
C. R Package Requirements and Installation	20	
D. Dataset Description and Access	15	
E. Content Quality and Concept Explanations	25	
F. R Code Quality and Annotation	25	
G. Structure, Organization, and Flow	10	
H. References and Attribution	10	
I. Professionalism and Submission Requirements	5	
TOTAL	150	

4. Feedback Summary

Strengths

Areas for Improvement

Final Comments